Raspberry Pi microcontroller chips

Source: https://github.com/raspberrypi/documentation/blob/7376533ab86ea07acc77ef6a9f3c3e57fcff833d/documentation/asciidoc/microcontrollers/microcontroller-chips.adoc (CC BY-SA 4.0; adapted for this benchmark)

Raspberry Pi microcontroller chips

Raspberry Pi designs its own microcontrollers, which are standalone silicon chips for small embedded projects that provide real-time hardware control. These chips are used in Raspberry Pi Pico boards and can be used in third-party microcontroller boards. For more information about Raspberry Pi Pico, see Pico microcontroller boards.

Raspberry Pi microcontrollers can be programmed using:

C or {cpp}, which are performance focussed and provide full control over hardware.

MicroPython, which provides simpler scripting for rapid development.

For more information on developing with RP2040 or RP2350, see the C/{cpp} SDK and MicroPython documentation.

Microcontroller overview

Raspberry Pi microcontrollers are silicon chips that handle specific, real-time tasks, like controlling motors or reading sensors, in embedded systems. Raspberry Pi offers two microcontrollers:

RP2040 is the original, general-purpose, dual-core Raspberry Pi microcontroller: a simple, cost-effective microcontroller unit (MCU).

RP2350 is a high-performance, dual-core microcontroller for applications with flexible hardware control; it's a more powerful MCU with advanced features, memory, and security. RP2350 has four variants, described in RP2350 variants.

Key features

Both RP2040 and RP2350 share the following core features:

Dual-core processors for parallel processing.

On-chip SRAM for real-time workloads.

Deterministic bus fabric through a multi-master AHB crossbar that connects the CPUs, DMA, and peripherals to multi-bank SRAM.

Built-in USB controller and PHY that allows the microcontroller to act as either a USB device or USB host.

Flexible general-purpose I/O (GPIO) that can be configured for digital input/output or assigned to peripheral functions like UART, SPI, I2C, or PWM.

Programmable I/O (PIO), a unique feature of Raspberry Pi microcontrollers that provides extended peripheral support.

Built-in hardware peripherals for interfaces such as UART, SPI, I2C, PWM, and so on, reducing software overhead and improving efficiency.

High performance, low power consumption, and low power leakage as a result of the modern 40 nm manufacturing process for microcontrollers.

Low-power modes to support extended-duration operation on battery power.

Programming options with C/{cpp} SDK or MicroPython; you can run Python through the MicroPython port and program it using the UF2 bootloader built into ROM for drag-and-drop programming.

Feature comparison

The following table provides a comparison between RP2040 and RP2350 microcontrollers. For more details about each microcontroller's features, see the dedicated sections for RP2040 and RP2350.

.Comparison of Raspberry Pi microcontrollers (RP2040 and RP2350)

Feature | RP2040 | RP2350

Type

General-purpose MCU

High-performance MCU

Processor core

Dual-core Arm Cortex-M0+

Dual-core Arm Cortex-M33 (higher performance, TrustZone support) or Hazard3 (RISC-V)

Clock speed

Up to 133 MHz

Up to 150 MHz

On-chip SRAM

264 kB in six independent banks

520 kB multi-bank performance

Flash (non-volatile) memory

No on-chip flash; support for up to 16 MB of off-chip flash memory through dedicated external (QSPI.XIP) bus.

Option for on-chip flash depending on variant; support for up to 32 MB of off-chip flash memory through dedicated external (QSPI.XIP) bus. See RP2350 variants.

External volatile working memory (PSRAM)

Not supported.

Supported by a QMI memory interface using the QSPI.XIP bus.

GPIO pins

30 GPIO pins, 4 of which can be used as analogue inputs.

Depending on variant: 30 GPIO pins, 4 of which can be used as analogue inputs; or 48 GPIO pins, 8 of which can be used as analogue inputs. See RP2350 variants.

Programmable Input/Output (PIO)

Two PIO blocks, each with four state machines (eight in total).

Three PIO blocks, each with four state machines (twelve in total).

Peripherals

See RP2040 features.

Expanded set; see RP2350 features.

Direct Memory Access (DMA) controller

12-channel DMA controller with peripheral pacing and chaining.

16-channel DMA with enhanced throughput and flexibility.

AHB crossbar

A fully connected, multi-master AHB-Lite crossbar connecting CPUs, DMA, and peripherals to multi-bank SRAM and flash.

AHB5 crossbar, offering higher-throughput and bandwidth.

Interpolator

Dual interpolator blocks for fast integer maths and table lookup.

Dual interpolator blocks for fast integer maths and table lookup.

Integer divider

Separate hardware integer divider accessed through memory-mapped register.

Not present because the processors support hardware division in the CPU.

Analogue-to-digital conversion (ADC)

4 analogue inputs with internal temperature sensor, 500 kS/s, and 12-bit conversion.

Depending on variant: 4 or 8 analogue inputs with internal temperature sensor, 500 kS/s, and 12-bit conversion.

Core regulator architecture

On-chip, programmable Low-Dropout (LDO) to generate core voltage.

On-chip buck converter (SMPS) for active operation plus optional low-quiescent-current programmable LDO mode for sleep states.

On-chip Phase-Locked Loops (PLL)

Two PLLs for core clock and USB clock generation.

Two PLLs for core clock and USB clock generation.

Security features

None (basic MCU).

Arm TrustZone; cryptographic accelerators.

Deployment boards

Raspberry Pi Pico; third-party RP2040-based boards.

Raspberry Pi Pico 2; third-party RP2350-based boards.

Programming

C/{cpp} SDK, MicroPython

C/{cpp} SDK, MicroPython

USB identifiers

Many RP2040 and RP2350 chips use a combination of Raspberry Pi USB identifiers (Vendor ID and Product ID). If you make your own, third-party board based on RP2040 or RP2350, you might need a unique USB Product ID (PID) if:

You need to provide a custom Windows driver.

You want your device to be uniquely identifiable as a separate product.

Raspberry Pi has permission from USB-IF to licence USB PID values for its Vendor ID (0x2E8A) for common silicon components used with RP2040 and RP2350. This means that you can assign unique PIDs under our Vendor ID (0x2E8A). To reserve a USB PID associated with Raspberry Pi's Vendor ID, follow the instructions in the Raspberry Pi USB PID Git repository.

If you don't need a unique PID, you can still uniquely distinguish devices using any of the following USB text fields: iManufacturer, iProduct, and iSerial.